T3 B_+1: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

B_+1

Additional quartics: 1, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

General-interaction coupling index

Coupling	Field content / definition	Contraction / source

$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_1)^2$	Contraction $I = 1$; exact tensor in registry `lambdaT3GeneralQ022000I1`

Each X_i labels one field-content class in the general-quartic registry; a superscript distinguishes different invariant contractions where present. The original RGBeta identifier and contraction are retained in the index. This is notation only, not a physical normalisation mapping.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{17{g_1}^3}{2}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{17{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ & + \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ & - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ & - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & -\frac{9{g_2}^2 m_1^2}{2} \\ & - \frac{3{g_1}^2 m_1^2}{2} \\ & + 4\lambda_{12}^{(1)} m_2^2 \\ & + 6\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & -\frac{9{g_2}^2 m_2^2}{2} \\ & - \frac{27{g_1}^2 m_2^2}{2} \\ & + 4\lambda_{12}^{(1)} m_1^2 \\ & + 6\lambda_{S_2}^{(1)} m_2^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\,(\lambda_5)^* \\ & + 8\lambda_{X_{10}}\,(\lambda_{X_{10}})^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1 \\ & + \frac{3{g_1}^4}{4}\end{aligned}$$

$$
\begin{aligned}
&+ 12{\lambda_1}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & 8\lambda_{X_{10}} \left(\lambda_{X_{10}}\right)^* \\
& + \frac{9{g_2}^4}{4} \\
& + {g_1}^2 \left(\frac{3{g_2}^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
& - 9{g_2}^2\lambda_{S_1}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + \frac{{\lambda_{12}^{(A)}}^2}{4} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
& + 12{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \frac{9{g_2}^4}{4} \\
& + {g_1}^2 \left(\frac{27{g_2}^2}{2} - 27\lambda_{S_2}^{(1)}\right) \\
& - 9{g_2}^2\lambda_{S_2}^{(1)} \\
& + \frac{243{g_1}^4}{4} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + \frac{{\lambda_{12}^{(A)}}^2}{4} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + \frac{{\lambda_{HS_2}^{(A)}}^2}{4} \\
& + 12{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = \lambda_{HS_1}^{(1)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 6\lambda_5\,(\lambda_5)^* \\
&+ 24\lambda_{X_{10}}\,(\lambda_{X_{10}})^* \\
&+ \frac{9{g_2}^4}{4} \\
&- 9{g_2}^2\lambda_{HS_1}^{(1)} \\
&+ \frac{3{g_1}^4}{4} \\
&- 3{g_1}^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \frac{3{\lambda_{HS_1}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = \lambda_{HS_2}^{(1)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 6\lambda_5\,(\lambda_5)^* \\
&+ \frac{9{g_2}^4}{4} \\
&- 9{g_2}^2\lambda_{HS_2}^{(1)} \\
&+ \frac{27{g_1}^4}{4} \\
&- 15{g_1}^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3{\lambda_{HS_2}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = &3\lambda_5\,(\lambda_5)^* \\
&+ \frac{9{g_2}^4}{4} \\
&- 9{g_2}^2\lambda_{12}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{27{g_1}^4}{4} \\
&- 15{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5 \Bigg(& 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \Bigg) \\
& - 9{g_2}^2\lambda_5 \\
& - 9{g_1}^2\lambda_5 \\
& + 2\lambda_{12}^{(1)}\lambda_5 \\
& + 2\lambda_1\lambda_5 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_5 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)} (& 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + \frac{16}{3}\lambda_5\,(\lambda_5)^* \\
& + 64\lambda_{X_{10}}\,(\lambda_{X_{10}})^* \\
& + {g_1}^2\left(6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\right) \\
& - 9{g_2}^2\lambda_{HS_1}^{(A)} \\
& + \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
& + 2\lambda_1\lambda_{HS_1}^{(A)} \\
& + 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
& + 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)} (& 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + 16\lambda_5\,(\lambda_5)^* \\
& + {g_1}^2\left(18{g_2}^2 - 15\lambda_{HS_2}^{(A)}\right) \\
& - 9{g_2}^2\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1 \lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)} \lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)} \lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = &\frac{4}{3}\lambda_5 \left(\lambda_5\right)^* \\
&+ {g_1}^2 \left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right) \\
&- 9{g_2}^2 \lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)} \lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)} \lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{X_{10}}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_{10}}}{d\ln\mu} = &\lambda_{X_{10}} (6\,\mathrm{Tr}\Big(Y_d\ \left(Y_d\right)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ \left(Y_e\right)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ \left(Y_u\right)^\dagger\Big)) \\
&- 9{g_2}^2 \lambda_{X_{10}} \\
&- 3{g_1}^2 \lambda_{X_{10}} \\
&+ 2\lambda_1 \lambda_{X_{10}} \\
&+ 8\lambda_{HS_1}^{(1)} \lambda_{X_{10}} \\
&+ 4\lambda_{HS_1}^{(A)} \lambda_{X_{10}} \\
&+ 2\lambda_{S_1}^{(1)} \lambda_{X_{10}}
\end{aligned}
$$